Practice Problems 4: Solutions

ECON 441: Introduction to Mathematical Economics | Instructor: Div Bhagia

Exercise 4.4

5. (e)

$$A = \begin{bmatrix} -2 \\ 4 \\ 7 \end{bmatrix}_{3\times1} \qquad B = \begin{bmatrix} 3 & 6 & -2 \end{bmatrix}_{1\times3}$$

$$C = AB = \begin{bmatrix} -6 & -12 & 4 \\ 12 & 24 & -8 \\ 21 & 42 & -14 \end{bmatrix}$$

$$D = BA = [3 \times -2 + 6 \times 4 + -2 \times 7] = [4]$$

7. In example 5, $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $A = \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix}$. In which case,

$$\begin{aligned} x'Ax &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2\times1} \\ &= \begin{bmatrix} a_{11}x_1 & a_{22}x_2 \end{bmatrix}_{1\times2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2\times1} \\ &= \underbrace{a_{11}x_1^2 + a_{22}x_2^2}_{\text{Weighted sum of squares}} = \sum_{i=1}^{2} a_{ii}x_i^2 \end{aligned}$$

So $x'Ax$ represents a weighted sum of squares where a_{11}, a_{22} are weights.

But now what if $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$. In this case,

$$x'Ax = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}_{2\times2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2\times1}$$

$$
\begin{aligned}
&= \begin{bmatrix} a_{11}x_1 + a_{21}x_2 & a_{12}x_1 + a_{22}x_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2\times 1} \\
&= a_{11}x_1^2 + a_{21}x_1x_2 + a_{12}x_1x_2 + a_{22}x_2^2 \\
&= a_{11}x_1^2 + (a_{21} + a_{12})\, x_1x_2 + a_{22}x_2^2
\end{aligned}
$$

So $x'Ax$ no longer represents a weighted sum of squares.

You can check that the associative law i.e.

$$(x'A)\, x = x'(Ax)$$

will apply in both cases (after all, its a law!) as all products are possible.

Exercise 4.5

1.

$$A = \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix}_{2\times 3} \qquad B = \begin{bmatrix} 9 \\ 6 \\ 0 \end{bmatrix}_{3\times 1} \qquad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2\times 1}$$

(a) $$AI = \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3} = \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix} = A$$

(b) $$IA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times 2} \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix} = \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix}$$

(c) $$Ix = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times 2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

(d) $$x'I = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times 2} = \begin{bmatrix} x_1 & x_2 \end{bmatrix}$$

4. Let's start with a 2×2 diagonal matrix

$$\begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix} \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix} = \begin{bmatrix} a_{11}^2 & 0 \\ 0 & a_{22}^2 \end{bmatrix}$$

$x = x^2$ for only $x = 0, 1$ so a_{11} and a_{22} can either be 0 or 1 . So we can have the following 2×2 idempotent diagonal matrices:

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

More generally, for $n \times n$ matrix, there can be 2^n such matrices. This is because there will be n elements, each of which can take two values.

Exercise 5.1

3.

a) Yes

b) Yes

c) Yes

d) No, the second row = 2 $\times$ first row.

4. Yes, we get the same answer.